

AKHIL KUMAR MADINENI

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Portfolio: akhii.vercel.app

PROFESSIONAL SUMMARY

AI Engineer with over 1.5 years of experience building production large language model (LLM) applications, retrieval-augmented generation (RAG) pipelines, and agent-based AI workflows. Accelerated enterprise automation by 35% through multi-agent system design. Delivered a 40% uplift in answer precision across enterprise knowledge bases via RAG-driven solutions. Expertise in LangChain, LangGraph, Temporal, vector databases, and SOC 2 compliance. Seeking an AI Engineer role to scale intelligent systems.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

AI & Machine Learning: LangChain (AI framework), LangGraph (agent orchestration), LangSmith (observability), OpenAI API, RAG Pipelines, AI Agents, Vector Databases (Qdrant, ChromaDB), Prompt Engineering

Backend & DevOps: FastAPI, Temporal (workflow engine), Spring Boot, Docker, Kubernetes, AWS (EC2, S3, CodeArtifact), CI/CD (Jenkins, GitHub Actions), Prometheus, Grafana

Databases & ETL: PostgreSQL, MySQL, MariaDB, Redshift, Pentaho, Prisma ORM, Hibernate

EXPERIENCE

AI Engineer & Backend Engineer

Aunix AI | Hyderabad, Telangana | July 2025 – Present

- Architect multi-agent AI systems using LangGraph (agent orchestration framework), reducing workflow completion time by 35% across 3 enterprise deployments.
- Develop production retrieval-augmented generation (RAG) pipelines with LangChain, OpenAI, and Qdrant vector database, improving knowledge retrieval accuracy by 40%.
- Engineer Pentaho ETL plugins for MariaDB, Redshift, NetSuite, and SharePoint, integrating 20+ enterprise data sources.
- Deploy Prometheus and Grafana monitoring dashboards, cutting client-reported incidents by 30%.
- Strengthen SOC 2 compliance through role-based access control (RBAC), audit logging, and data access policies.
- Publish custom Docker images for Temporal (workflow engine) via Jenkins and GitHub Actions CI/CD; distribute packages to AWS CodeArtifact (artifact repository).

Key Projects at Aunix AI:

- Aunix Platform** (Python, LangGraph, LangChain, Temporal, OpenAI, ChromaDB, RustFS object storage)
 - Planned and built AI agent workflows for enterprise automation and multi-step reasoning, serving 100+ concurrent users.
 - Constructed scalable RAG pipelines for enterprise knowledge management, reducing manual query time by 50%.
 - Executed fault-tolerant distributed workflows with Temporal and intelligent LLM routing, achieving 99.5% task completion rate.
 - Introduced an index-based RAG agent using Page Index for unstructured data retrieval, cutting indexing overhead by 60%.
- Flodata ETL Platform** (Java, Spring Boot, Pentaho, MariaDB, Redshift, NetSuite)
 - Delivered enterprise-grade ETL plugins for Flodata, Aunix's manual flow-building data platform, supporting ingestion of 5M+ records per run with load times under one hour.

- Created monitoring dashboards and audit logging pipelines, improving platform reliability and reducing alert response time by 40%.
- Resolved SOC 2 compliance gaps through stronger access controls and comprehensive audit trails, passing third-party review.

Software Engineer Trainee

Silicon Techlab Private Limited | Bhubaneswar, Odisha | January 2025 – July 2025

- Implemented full-stack user management with JWT authentication, supporting 100+ internal users with zero reported authentication failures.
- Modeled database schemas and optimized SQL queries for student records, reducing average query latency by 25%.
- Administered production databases, maintaining data integrity and validating backup and recovery procedures for 3 critical systems.

PROJECTS

Structly – System Design Workspace

Next.js, React 19, TypeScript, Excalidraw, Prisma ORM, NextAuth v5

- Cloud-first workspace for documenting and visualizing distributed system architectures, used by 200+ engineers.
- Integrated Excalidraw Canvas API and developed a diagram parser converting Mermaid syntax to Excalidraw format, reducing diagram setup time by 70%.
- Incorporated rich-text architectural notes with a collaborative text editor and Markdown support for structured documentation.
- Configured NextAuth v5 with Prisma ORM for GitHub and Google OAuth, ensuring complete per-user data isolation.
- Live Demo: structly.free-tools.space

Chords Music Player

Java, Spring Boot, React, JavaScript, REST APIs

- Developed a full-stack, ad-free music streaming application with playlist management and real-time audio streaming for 500+ tracks.
- Received 50+ open-source community contributions and positive feedback within 3 months of launch.
- Live Demo: human-music.vercel.app

EDUCATION

Bachelor of Technology in Computer Science and Engineering (Computer Networking Specialization)

Centurion University of Technology and Management, Vizianagaram, Andhra Pradesh | 2021 – 2025 | GPA: 9.0/10.0

CERTIFICATIONS

- LangChain for LLM Application Development – DeepLearning.AI (In Progress)
- AWS Certified AI Practitioner – Scheduled